

Guiding Mixture-of-Experts Reasoning via Routing Intervention

1 Introduction

2 Background

3 Related Work

4 Routing Intervention

5 Experimental Setup

Current experimental scope. All results available at the time of writing use `allenai/OLMoE-1B-7B-0924-Instruct` (16 MoE layers, 64 experts, $\text{top-}k = 8$). The current successful runs are GSM8K (1,319 generated traces), GSM8K with a self-check prompt (1,319), MATH (1,500), MATH with a self-check prompt (1,500), and ProcessBench (400 given-solution traces). All five have complete router and hidden-state tensors and outputs for Experiments 1–5. The current PRM800K launch is not a completed experiment: its Experiment 1 JSONL is empty, and downstream stages produced zero-input placeholder files. Older, corrected-label PRM800K event and full-trace probe outputs over 10,833 traces are retained as explicitly marked legacy evidence. Unless stated otherwise, event windows contain five tokens on each side of the event. In addition, Experiment 0a is an auxiliary judge-validation study: completed static-reward baselines optimize a Qwen3.6-27B prompt for sentence-level Schoenfeld episode classification, while a new LLM-as-a-judge reward configuration awaits rerunning. This judge experiment evaluates annotation capability and is not evidence about Qwen router dynamics. Experiment 0b is a separate representation-probe replication: it teacher-forces the 38 released DeepSeek-R1 Schoenfeld traces through the official GPTQ 4-bit Qwen/Qwen3.5-35B-A3B-GPTQ-Int4 checkpoint and asks whether the label of the next sentence is linearly decodable at its causal pre-sentence boundary. It is an auxiliary hidden-state experiment rather than a routing result from the OLMoE pipeline.

Implemented analysis pipeline. The repository implements trace generation or ingestion (Experiment 1), token-level router and optional hidden-state extraction (Experiment 2), event-centered routing with matched within-trace controls, geometry–routing correlation (Experiment 3), prospective prefix probes (Experiment 4), and expert phase/co-activation/transition analysis (Experiment 5). Offline tools also support high-precision relabelling, rebuilding multi-dataset taxonomy summaries, full-trace layerwise linear probes over router, hidden, or combined features, and a trace-length/structure baseline. The full pipeline runs Experiments 1–5 and extracts hidden states by default; the latter and Experiment 4 can be disabled together when storage is constrained. Experiment 0a is run separately: a local llama.cpp server supplies Qwen3.6-27B predictions while GEPA optimizes the instruction. The current version supplies the problem statement and previous/current/next response units, uses 21 audited synthetic contrastive demonstrations, supports an informative same-model LLM-as-a-judge reward alongside the earlier static rewards, and evaluates prompt-selection rules with nested response-grouped cross-validation.

Experiment 0b separately aligns the released sentence annotations to their original responses, extracts one 2,048-dimensional boundary vector at each of 41 hidden-state indices, trains layerwise episode probes, and applies

Author’s Contact Information:



This work is licensed under a [Creative Commons Attribution International 4.0 License](https://creativecommons.org/licenses/by/4.0/).

© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM.

DOI:

their selected layers to reference or supplied solutions from four external math benchmarks. Only the boundary vectors are stored; Qwen does not regenerate the released or benchmark solutions.

Prospective probe design and status. Experiment 4 pools only the first 25%, 50%, or 75% of each trace and trains separate trace-grouped logistic probes from structure-only, router-only, hidden-only, and combined hidden/router features. It evaluates final incorrectness and whether a gold first error remains beyond the observed prefix, excluding traces whose error has already occurred for the latter target. Outputs include out-of-fold AUROC, balanced accuracy, F1, and trace-bootstrap confidence intervals. This pipeline is implemented and has completed for all five current conditions. Final-incorrectness probes are available everywhere. Future-first-error probes are available only for ProcessBench because the generated datasets do not have gold error-step annotations.

Schoenfeld boundary-probe design. Experiment 0b starts from 3,125 annotated units in 38 released DeepSeek-R1 SAT responses. It excludes the 38 synthetic `</think>` closing units, which are all labelled Monitor, leaving the paper-matched 3,087 units. Qwen receives each SAT instruction as the user message and the released response as a teacher-forced assistant continuation. For a target unit, the feature is the hidden state immediately before its first token, so the probe sees the problem and preceding response but cannot read the target sentence. Hidden-state index 0 is the embedding output and indices 1–40 are the successive language-model layers.

At each index, seven separate target-versus-rest logistic regressions use raw activations, balanced class weights, L2 regularization with $C = 1$, and no standardization or PCA. The paper-matched protocol uses a target-stratified 80/20 sentence split with seed 42. This is a replication choice rather than a strong generalization design: all 38 response IDs occur in both partitions. The reported layer for each target maximizes accuracy on that same test split. For qualitative transfer inspection, the selected probes are then applied to 20 examples each from GSM8K, MATH, PRM800K, and ProcessBench. The seven binary scores are independently trained and uncalibrated; their argmax is an inspection label, not a multiclass probability or new gold annotation.

Important provenance note. The event-label heuristic was made substantially more conservative after the first experimental run. Taxonomy, event-routing, and expert-event outputs for the five current conditions were regenerated with the corrected labels, and their JSONL files pass line-by-line JSON validation. The separate PRM800K relabelled files and full-trace probe outputs predate the current pipeline run and are never pooled with current estimates. SLURM’s printed “Pipeline complete” banner is not treated as evidence of success: both PRM800K logs contain explicit zero-input errors even though the shell script continued to that banner. Historical logs also include a CUDA out-of-memory failure and several OLMoE/Qwen MoE-generation logs that end mid-progress; no Qwen routing-result artifacts are present, so these are excluded rather than counted as robustness runs. The separate Qwen judge artifacts from Experiment 0a and the Qwen3.5 boundary-probe artifacts from Experiment 0b are complete. Repeated cgroup swap-limit warnings also appear in the successful jobs and are treated as environmental warnings, not failures.

6 Results

6.1 GEPA-optimized episode judge

Experiment 0a evaluates whether prompt optimization improves a local judge for the seven adapted Schoenfeld episode labels: Read, Analyze, Plan, Implement, Explore, Verify, and Monitor. The model was the Q4_K_XL quantization of Qwen3.6-27B served by llama.cpp on one RTX 3090. All splits preserve complete responses, preventing neighboring units from crossing partitions.

Preliminary single-split runs. With seed 42, the 38 responses were divided into 26 training, 6 validation, and 6 test documents (2,382/407/336 units). These first runs classified the current unit without neighboring context and optimized per-unit exact match.

Table 1. Preliminary Experiment 0a judge agreement. The composite rescales Cohen’s κ and Kendall’s τ_b from $[-1, 1]$ to $[0, 1]$ and averages them. Coverage was 100% in every row.

Evaluation	Accuracy	κ	τ_b	Composite
Seed validation	0.6388	0.5581	0.4810	0.7598
Optimized validation	0.6855	0.6074	0.5847	0.7980
Optimized test	0.6607	0.5710	0.6066	0.7944
Few-shot validation	0.6806	0.6078	0.5830	0.7977
Few-shot test	0.6994	0.6178	0.6667	0.8211

The matched few-shot run used the same seed, document split, model, and GEPA budget, adding seven gold training examples to the instruction. It performs better than the optimized base condition on the held-out test set: accuracy rises from 66.07% to 69.94% (+3.87 percentage points), Cohen’s κ from 0.5710 to 0.6178, Kendall’s τ_b from 0.6066 to 0.6667, and composite agreement from 0.7944 to 0.8211. GEPA itself does not improve the few-shot validation result: the seed and selected optimized prompts both score 68.06% accuracy. The current comparison therefore favors few-shot prompting for held-out generalization, whereas the measurable GEPA optimization gain is confined to the base-prompt condition.

Context-aware nested cross-validation. The revised protocol excludes the same six responses from model selection and runs five outer folds over the remaining 32 responses. Within each fold, five responses form the inner validation set and all others outside the outer fold form the GEPA training set. Each request includes the problem statement, previous unit, current unit, and next unit. The seed instruction includes 21 audited synthetic contrastive demonstrations. GEPA receives inverse-frequency weighted exact match, whose split mean is balanced accuracy; the safety selector compares either balanced accuracy or macro-F1 and rejects any candidate whose class recall falls by more than the configured threshold.

New informative reward (implementation complete; run pending). The current implementation replaces the default static GEPA reward with an LLM-as-a-judge call to the same Qwen3.6-27B model. For each candidate label, the judge returns a short free-text explanation plus four diagnostic scores from 1 to 5: gold agreement, functional fit, contextual coherence, and boundary precision. Their normalized weighted mean is used only as GEPA’s optimization signal, with respective weights 0.40, 0.25, 0.15, and 0.20. Gold-label accuracy, balanced accuracy, macro-F1, and per-class recall remain the independent selection and reporting criteria. The judge currently runs deterministically with model reasoning disabled; reasoning-enabled evaluation is deferred to a 5090 run because it substantially increases inference time. Every judge call is persisted as JSONL with its component scores, normalized score, and free-text explanation, permitting a direct audit of feedback quality. The pending rerun uses GEPA’s heavy budget and a 4096-token reflection cap while retaining the existing one-GPU, 64 GB, 48-hour SLURM allocation. All numerical results below precede this change and therefore remain static-reward baselines rather than evidence for the new judge.

Macro-F1 selection with the light budget is the strongest cross-validated configuration, but its advantage over balanced-accuracy selection is small: +0.32 accuracy points, +0.43 balanced-accuracy points, and +0.24 macro-F1 points. At the response level it wins on 15 documents, loses on 13, and ties on 4, so the observed gain is not stable across documents. The light run selects a GEPA-optimized prompt in three of five folds; the recall gate retains the seed prompt in the other two because Plan or Monitor recall falls too far.

Increasing the budget does not help. The heavy run selects an optimized prompt in only two folds and is 0.65 accuracy points and 0.68 macro-F1 points below the light run. Paired predictions show 38 units correct only under light versus 20 correct only under heavy. This difference is concentrated in fold 5, where light discovers

Table 2. Aggregate selected-prompt outer-fold results over 2,789 units. “Balanced light” is the earlier 0.10 recall-gate run; the macro-F1 runs use a 0.05 gate. The six excluded responses were not evaluated.

Selection / budget	Accuracy	Balanced acc.	Macro-F1	Composite
Balanced / light	0.6701	0.6753	0.6480	0.8031
Macro-F1 / light	0.6734	0.6796	0.6505	0.8063
Macro-F1 / heavy	0.6669	0.6737	0.6437	0.8030

Table 3. Per-class outer-fold performance for the best current configuration, macro-F1 selection with the light GEPA budget.

Class	Support	Precision	Recall	F1
Read	318	0.798	0.770	0.784
Analyze	685	0.757	0.632	0.689
Plan	205	0.524	0.590	0.555
Implement	650	0.880	0.722	0.793
Explore	193	0.679	0.492	0.571
Verify	538	0.669	0.606	0.636
Monitor	200	0.364	0.945	0.526

and selects a strong candidate whereas heavy retains the seed after its candidate violates the Analyze recall gate. Independent heavy search is therefore not a superset of light search and can discard a better prompt found by a smaller run.

The revised objective improves overall balance only marginally and does not resolve Monitor overprediction. The model emits Monitor 519 times for 200 gold Monitor units; the leading errors are Verify→Monitor (127), Plan→Monitor (65), and Explore→Monitor (42). The annotation audit flags 180/3,125 units (5.76%), including 145 multiline units and 49 structural/control markers such as `</think>` and “Final Answer.” These artifacts plausibly inflate Monitor recall while depressing precision and should be reviewed before further optimization.

All runs retain 100% final-output coverage and complete without OOM failures, but the light and heavy jobs log 31 and 28 truncated LM responses during optimization. Moreover, the outer folds currently evaluate only the safety-selected prompt, not both seed and optimized prompts, so they do not directly estimate GEPA’s paired outer-fold gain. Finally, the six responses excluded from current CV were inspected in the preliminary experiments; they are locked from the current code path but no longer constitute a pristine research holdout. The judge should therefore not replace validated annotation until structural artifacts are adjudicated, decoding is constrained, paired outer evaluation is added, and a genuinely unseen external test set is designated.

6.2 Schoenfeld episode-boundary probes

Experiment 0b tests a different annotation mechanism from the generative judge in Experiment 0a. It asks whether Qwen’s representation of the prefix immediately before a sentence contains linearly accessible information about the upcoming Schoenfeld episode. It therefore measures prospective prefix-state decodability, not semantic classification after reading the sentence and not the latent dynamics of a Qwen-generated trace.

Within-corpus results. Table 4 reports the accuracy-selected layer for each one-versus-rest target. AUROC is 0.835–0.934 and balanced accuracy is 0.740–0.862, demonstrating substantial within-split ranking signal. Implement is strongest at the natural threshold (F1 0.771), followed by Verify (0.726) and Read (0.704). Analyze

Table 4. Accuracy-selected Schoenfeld boundary probes on the sentence-level test split. Prevalence is over all 3,087 units. Δ Maj. subtracts the always-negative accuracy. Balanced accuracy is derived from the saved test confusion matrix. Layer selection and evaluation use the same test split and are therefore selection-optimistic.

Target	Prevalence	Layer	Accuracy	Δ Maj.	Balanced acc.	F1	AUROC
Read	12.21%	34	0.9239	+0.0461	0.8476	0.7044	0.9338
Analyze	25.62%	36	0.8107	+0.0669	0.7669	0.6465	0.8348
Plan	7.29%	39	0.9175	-0.0096	0.7405	0.4848	0.8551
Implement	21.87%	38	0.8964	+0.1151	0.8617	0.7714	0.9305
Explore	6.74%	40	0.9223	-0.0103	0.7486	0.4894	0.8884
Verify	20.38%	27	0.8754	+0.0792	0.8509	0.7260	0.9122
Monitor	5.90%	35	0.9304	-0.0106	0.7546	0.4819	0.9031

is intermediate. Plan, Explore, and Monitor have F1 below 0.50. Their nominal accuracies are also 0.96–1.06 percentage points below an always-negative majority classifier, so accuracy alone gives a misleading impression for these rare targets.

The signal emerges rapidly and is not maximal at the model output. Mean one-versus-rest AUROC across targets rises from 0.624 at the embedding output to 0.873 at layer 10 and 0.892 at layer 20, peaks at 0.899 at layer 30, and falls to 0.867 at layer 40. Mean F1 peaks at layer 28. Accuracy-based target selection sometimes obscures this profile: for Plan, the selected layer 39 has AUROC 0.855 whereas layer 21 reaches 0.922. AUROC or balanced accuracy, chosen inside a validation fold, better matches the scientific question of decodability.

The fits are numerically stable—none of the 287 regressions reports a convergence warning—but statistical overfitting remains visible. At the selected layers, train–test accuracy gaps are 16.7 points for Analyze, 10.1 for Implement, approximately 7.8 for Plan and Explore, and 4.5–5.5 for the other targets. Explore reaches 100% training accuracy. With 2,048 features, 2,469 training units, strong within-response dependence, and only 38 distinct responses, the effective independent sample size is much closer to the number of responses than to the number of sentences. Test-based layer selection adds further optimism.

External benchmark inspection. All 80 requested examples complete without omitted units, yielding 581 sentence boundaries. Table 5 shows that the seven-way inspection behavior collapses mainly to Implement, Read, and Monitor. Implement alone receives 49.2% of argmax assignments, whereas Explore is never selected. The transfer distribution also differs sharply from training: Analyze falls from 25.6% of gold units to 7.7% of predictions and Verify from 20.4% to 3.3%.

The independent binary scores do not form a clean multiclass decision rule. Seventy-four units (12.7%) have no score above 0.5 but are still assigned an argmax label, while 180 (31.0%) activate more than one probe and 99 (17.0%) have a top-two margin below 0.1. A conservative display rule requiring a top score of at least 0.8 and a margin of at least 0.2 retains only 318/581 units (54.7%), mostly Implement, Read, and Monitor. This threshold is a useful abstention heuristic, not a calibrated guarantee of correctness.

Pragmatically, Implement is the most coherent transferred label: it marks long calculation, substitution, and algebra runs. Read often marks givens or restatements, particularly near the start of MATH and PRM800K solutions. Monitor behaves primarily as a solution-boundary or opening-state detector: all 20 GSM8K first units and 18/20 ProcessBench first units receive Monitor, whereas 19/20 MATH and 17/20 PRM800K first units receive Read. Because the target sentence is unseen at this point, this source dependence must arise from the problem,

Table 5. Argmax inspection-label counts on unannotated external reasoning units. These are prediction frequencies, not accuracies; no compatible Schoenfeld labels are available for these examples.

Dataset	Units	Read	Analyze	Plan	Implement	Explore	Verify	Monitor
GSM8K	96	0	2	3	53	0	0	38
MATH	88	23	7	1	45	0	1	11
PRM800K	139	34	18	0	71	0	1	15
ProcessBench	258	69	18	12	117	0	17	25
Total	581	126	45	16	286	0	19	89

Table 6. High-precision trace taxonomy. Percentages are of all traces. SC denotes self-correction; the PRM800K row is a corrected legacy output rather than part of the current five-condition run.

Dataset	N	Accuracy	Backtracking	Contradiction	SC	Final-answer reversal
GSM8K	1,319	37.38%	1 (0.076%)	1 (0.076%)	1 (0.076%)	0 (0.000%)
GSM8K + self-check	1,319	36.01%	9 (0.682%)	3 (0.227%)	7 (0.531%)	0 (0.000%)
MATH	1,500	13.47%	0 (0.000%)	15 (1.000%)	0 (0.000%)	28 (1.867%)
MATH + self-check	1,500	11.80%	9 (0.600%)	19 (1.267%)	9 (0.600%)	40 (2.667%)
ProcessBench	400	48.25%	3 (0.750%)	0 (0.000%)	3 (0.750%)	1 (0.250%)
PRM800K (legacy)	10,833	60.17%	2 (0.018%)	87 (0.803%)	1 (0.009%)	10 (0.092%)

prefix position, and formatting rather than the displayed unit’s semantics. Analyze and Plan appear plausibly on some explanatory or phase-change boundaries but have low coverage. Explore is unusable for external tagging in the present model.

Verify has a narrower but potentially useful role as a review or reconsideration flag. All 17 ProcessBench Verify predictions occur at or after the benchmark’s annotated first error, including six within one trace that repeatedly reinterprets the question before ending with the wrong sign. This is not an error detector: errors generally occur early, Verify may merely mark late position or reflective language, and no PRM800K post-error unit is labelled Verify. It can prioritize passages for inspection but cannot judge correctness.

The qualitative stage also exposes an input problem independent of the classifier. Thirteen units are split immediately after abbreviations such as “Mr.”, producing fragments including “Therefore, Mr.” and inconsistent scores. External use should first adopt abbreviation- and LaTeX-aware segmentation. With the current outputs, the defensible practical representation is a coarse three-phase view—setup (Read and opening Monitor), execution (Analyze, Plan, and Implement), and review (Verify)—plus an explicit unknown/ambiguous state. The seven raw argmax labels should not be used as automatic cognitive annotations.

6.3 Reasoning outcomes and event taxonomy

Table 6 is recomputed directly from the current trace JSONL files; the PRM800K row alone comes from the separately retained corrected-label output. Accuracy is final-answer accuracy for generated traces and the stored outcome label for given solutions. Explicit revision language remains rare, but self-check prompting increases its incidence: backtracking rises from 1 to 9 traces on GSM8K and from 0 to 9 on MATH. This does not translate into successful repair. All seven GSM8K-self-check and all nine MATH-self-check traces labelled as self-corrections end incorrect.

Table 7. Number of traces carrying each label before and after the high-precision relabelling.

Dataset	Version	Backtrack	Contradiction	Self-correction
GSM8K	old	10	5	9
	current	1	1	1
PRM800K	old	112	145	61
	current	2	87	1
ProcessBench	old	14	2	11
	current	3	0	3

Table 8. ProcessBench routing around the gold first-error step. Values are averaged across traces and layers.

Metric	Before	At first error	After
Routing entropy	3.80831	3.80624	3.80520
Expert-switch rate	0.60402	0.59066	0.58478
Top- k overlap	0.27355	0.28462	0.28982
Router margin	0.01986	0.01940	0.01953

Because baseline and self-check generations use the same problem IDs, we also compare correctness within problem. On GSM8K, self-checking changes 168 baseline-correct answers to incorrect and 150 baseline-incorrect answers to correct (exact McNemar $p = 0.340$). On the 1,499 MATH problems with determined correctness in both conditions, the corresponding counts are 107 and 82 ($p = 0.081$). Thus neither dataset provides evidence that the self-check prompt improves accuracy; its observed point estimate is lower by 1.37 and 1.67 percentage points, respectively. The increased event count should therefore be interpreted as elicited revision language, not effective self-correction.

The earlier relabelling audit in Table 7 remains an important warning: the original phrase matcher had many false positives. For example, GSM8K backtracking labels fell from 10 to 1, ProcessBench from 14 to 3, and PRM800K from 112 to 2. Claims based on the old labels are discarded. Even after deliberate self-check elicitation, current trace counts of 7–9 self-corrections are too small for broad claims.

6.4 Event-centered routing

The most reliable event-centered result currently available is the ProcessBench analysis at the gold first-error step ($n = 207$ traces with an error). Table 8 reports the raw trajectory. All changes are small. The matched within-trace control analysis in Table 9 confirms the null result: every 95% bootstrap confidence interval includes zero. Thus, with the present layer-averaged metrics, there is no evidence of a discontinuous routing change at the first erroneous step.

Plain GSM8K still has only two event-eligible traces, so its event effects are case studies. Self-checking increases GSM8K coverage to ten eligible traces, but every bootstrap interval for backtracking, self-correction, and the first detected event contains zero. Table 10 summarizes the better-powered generated-data comparisons. Two small MATH entropy effects are resolved: contradiction in the baseline condition and the first detected event under self-checking. In MATH self-check backtracking/ self-correction spans, the router margin falls. The latter labels identify the same nine traces and nearly the same spans, so they are not independent replications. None of these

Table 9. Matched-control effect at the ProcessBench first-error step. The effect is event change minus change at a matched normal position in the same trace; $n = 207$, 2,000 bootstrap samples.

Metric	Mean matched-control effect	95% bootstrap CI
Routing entropy	0.00378	[−0.00878, 0.01587]
Expert-switch rate	−0.00759	[−0.03295, 0.01876]
Top- k overlap	0.00712	[−0.01553, 0.02996]
Router margin	−0.000318	[−0.001506, 0.000844]

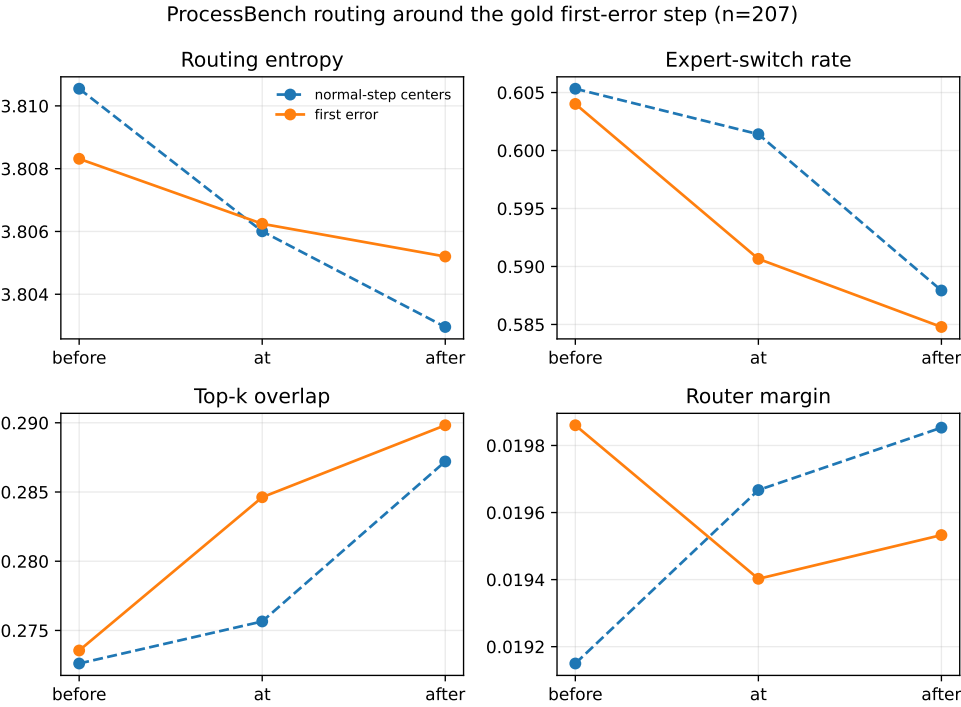


Fig. 1. Before/at/after trajectories for ProcessBench gold first errors and normal-step centers. The visual differences are descriptive; the controlled effects in Table 9 are not statistically resolved.

analyses corrects for the many event–metric comparisons, and trace counts are only 9–24; they are exploratory rather than evidence of a universal event signature.

The retained corrected PRM800K output contains 4,315 gold first-error traces, 87 traces with explicit contradictions, and 89 first heuristic reasoning events. Table 11 reports its earlier matched analysis. Contradiction and first-event intervals include zero. At the gold first error, switching falls, overlap rises, and margin rises by small amounts. It is a useful counterexample to a universal disruption hypothesis, but it is legacy evidence: the current PRM800K pipeline did not regenerate it.

Legacy PRM800K backtracking ($n = 2$) and self-correction ($n = 1$) remain case studies and are omitted from the inferential table.

Table 10. Current generated-data matched-control effects. Values are event change minus change at a matched normal position; intervals are trace-level 95% bootstrap intervals with 2,000 replicates.

Condition	Event	<i>n</i>	Δ entropy	Δ switch	Δ overlap	Δ margin
GSM8K + self-check	Backtracking	9	-0.00670 [-0.04040, 0.03064]	0.01237 [-0.06946, 0.09246]	-0.00226 [-0.07058, 0.06825]	-0.000833 [-0.004740, 0.003242]
GSM8K + self-check	Self-correction	7	-0.01062 [-0.05124, 0.03238]	0.03934 [-0.02679, 0.11244]	-0.02583 [-0.10064, 0.03729]	-0.000534 [-0.005338, 0.004480]
MATH	Contradiction	15	0.04381 [0.01039, 0.07702]	0.02031 [-0.06303, 0.10522]	-0.01834 [-0.09607, 0.06839]	-0.000884 [-0.005099, 0.003685]
MATH + self-check	First detected event	24	0.04130 [0.00639, 0.07399]	0.03743 [-0.03776, 0.10645]	-0.02995 [-0.09325, 0.03043]	-0.002898 [-0.005988, 0.000365]
MATH + self-check	Backtracking	9	0.03671 [-0.02635, 0.09093]	0.04991 [-0.04862, 0.13413]	-0.04524 [-0.11074, 0.03742]	-0.003962 [-0.007604, -0.000631]

Table 11. Legacy corrected PRM800K matched-control effects at the event. Intervals are trace-level 95% bootstrap intervals with 1,000 replicates.

Event	<i>n</i> traces	Δ entropy	Δ switch	Δ overlap	Δ margin
Contradiction	87	-0.00210 [-0.02381, 0.01722]	0.00576 [-0.03197, 0.04148]	-0.00324 [-0.03431, 0.02971]	0.000146 [-0.001789, 0.002247]
First reasoning event	89	-0.00491 [-0.02597, 0.01449]	0.00888 [-0.02478, 0.04288]	-0.00637 [-0.03866, 0.02663]	0.000511 [-0.001572, 0.002594]
Gold first error	4,315	-0.00187 [-0.00485, 0.00107]	-0.02120 [-0.02687, -0.01577]	0.01980 [0.01457, 0.02515]	0.000474 [0.000158, 0.000801]

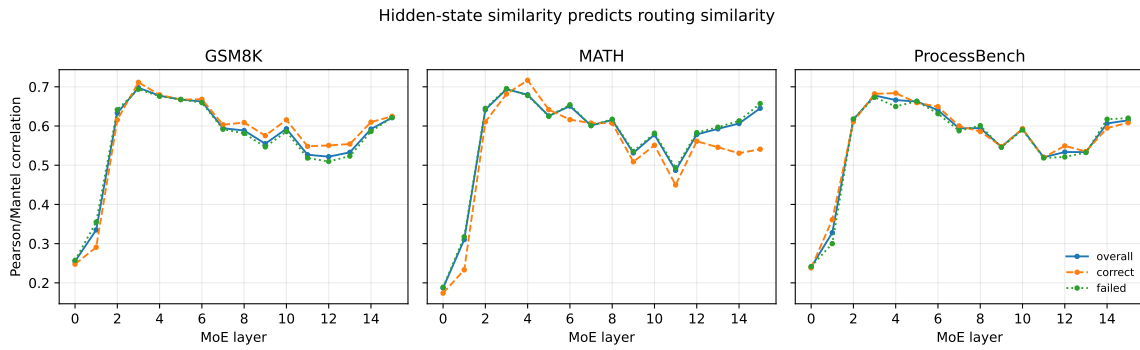


Fig. 2. Layerwise correlation between hidden-state similarity and routing- distribution similarity for three representative current conditions.

6.5 Hidden-state geometry and routing

For each layer, we compute the pairwise cosine-similarity matrix of sampled token hidden states and the cosine-similarity matrix of their softmax router distributions. Their upper triangles are correlated; the overall significance test is a 1,000-permutation Mantel test. This is a geometry- routing correlation analysis, not t-SNE and not a linear-separability probe. The exposed hidden state is the input to the transformer layer and is the closest available proxy for, but not exactly, the post-attention state seen by that layer’s router.

Figure 2 and Table 12 show the most replicated result in the study. Mean overall correlation lies in the narrow range 0.5605–0.5663 across all five current conditions. It rises sharply from layer 0 and peaks in layer 3 or 4 before declining moderately. Every layer in every condition reaches the permutation-resolution floor $p = 1/1001 \approx 0.000999$.

Correct-versus-failed differences are inconsistent. Correct traces have the higher mean on GSM8K, GSM8K self-check, and ProcessBench; failed traces have the higher mean on MATH and (slightly) MATH self-check. No trace-bootstrap interval is available for these contrasts. The overall analysis uniformly samples 1,000 tokens per condition and the Mantel permutation treats sampled tokens as exchangeable; it does not provide trace-level uncertainty or prove that representations cause routing. The defensible conclusion is narrower: hidden-state

Table 12. Summary of geometry–routing correlations across 16 layers.

Condition	Overall mean	Overall range	Peak layer	Correct mean	Failed mean
GSM8K	0.5659	0.2550–0.6977	3	0.5732	0.5635
GSM8K + self-check	0.5644	0.2529–0.6936	3	0.5830	0.5597
MATH	0.5641	0.1870–0.6940	3	0.5360	0.5676
MATH + self-check	0.5663	0.2194–0.6809	4	0.5655	0.5684
ProcessBench	0.5605	0.2399–0.6779	3	0.5637	0.5570

Table 13. Prospective final-incorrectness AUROC from prefix-only features. Representation entries select the best of 16 layers (shown in parentheses), so their comparison is exploratory and selection-optimistic. “Positive” is the number of finally incorrect traces.

Condition	Prefix	<i>N</i>	Positive	Structure	Router	Hidden	Combined
GSM8K	25%	1,319	826	0.6819	0.6512 (L10)	0.6559 (L12)	0.6514 (L10)
	75%	1,319	826	0.6819	0.6887 (L13)	0.6839 (L12)	0.6830 (L8)
GSM8K + self-check	25%	1,319	844	0.6960	0.6732 (L7)	0.6880 (L9)	0.6874 (L10)
	75%	1,319	844	0.6959	0.6843 (L9)	0.7070 (L10)	0.7105 (L10)
MATH	25%	1,499	1,297	0.7628	0.7144 (L12)	0.7189 (L14)	0.7252 (L14)
	75%	1,499	1,297	0.7628	0.7319 (L10)	0.7445 (L12)	0.7370 (L6)
MATH + self-check	25%	1,500	1,323	0.7736	0.7118 (L10)	0.7216 (L14)	0.7112 (L14)
	75%	1,500	1,323	0.7735	0.7446 (L10)	0.7420 (L15)	0.7402 (L15)
ProcessBench	25%	400	207	0.6192	0.6267 (L9)	0.6171 (L7)	0.6171 (L6)
	75%	400	207	0.6184	0.6685 (L4)	0.6931 (L11)	0.7022 (L11)

and router-distribution geometries are strongly aligned and the layerwise profile replicates across datasets and prompts, while outcome modulation does not.

6.6 Prospective prefix probes

Experiment 4 now supplies the prospective test that was previously missing. Table 13 reports early (25%) and late (75%) prefixes; the omitted 50% results lie between them except where noted below. On GSM8K and both MATH conditions, a three-feature length/step/character baseline is as good as or better than router, hidden, and combined representations at most prefixes. MATH is especially clear: structure AUROC is approximately 0.763–0.774 while the best representation probe reaches at most 0.745. This makes trace structure a major confound, not merely a weak baseline.

ProcessBench is the exception for final incorrectness: combined AUROC rises from 0.617 at 25% to 0.702 at 75%, versus a 0.618 structure baseline. The signal therefore emerges after much of the solution is visible, which is useful for monitoring but weaker evidence of advance prediction.

The stricter future-first-error target is available only on ProcessBench. At 25%, combined features reach AUROC 0.674 (95% bootstrap CI [0.612, 0.733]) versus 0.625 for structure. This advantage disappears at 50%, where structure is 0.640 and combined features are 0.563. At 75%, only 13 future errors remain; the best router AUROC is 0.682 with the very wide interval [0.535, 0.832]. Fractions condition on the error not having occurred

Table 14. Expert-activation shifts relative to normal tokens. Parentheses give phase-token counts; a dash indicates no tokens for that phase.

Dataset	Backtrack	Contradiction	Self-correction	First error	Final answer
GSM8K	0.0608 (61)	0.0804 (82)	0.0608 (61)	–	0.0136 (139,075)
GSM8K + self-check	0.0266 (2,512)	0.0390 (670)	0.0303 (1,997)	–	0.0129 (70,550)
MATH	–	0.0224 (1,677)	–	–	0.0160 (167,642)
MATH + self-check	0.0175 (2,345)	0.0154 (4,085)	0.0175 (2,345)	–	0.0143 (119,763)
ProcessBench	0.0592 (162)	–	0.0592 (162)	0.0104 (14,539)	0.0185 (13,210)

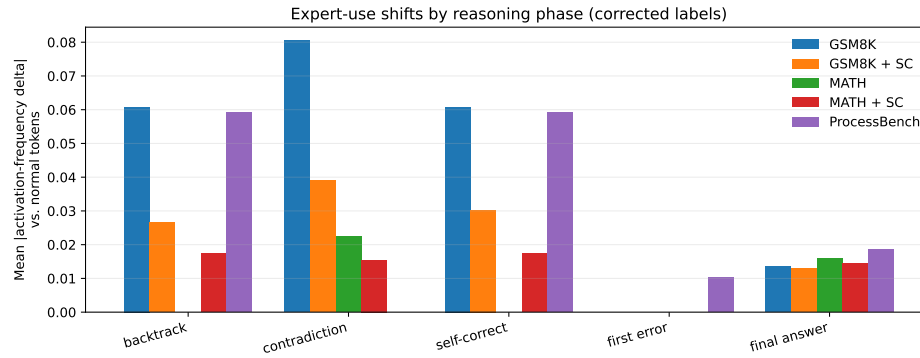


Fig. 3. Mean absolute expert-activation-frequency shift by reasoning phase. Explicit backtracking/self-correction bars have very small trace counts even when they contain many tokens.

yet, so their populations and class prevalence differ and should not be read as a learning curve. Together with best-layer selection and the absence of paired model-comparison tests, these results do not establish a robust advance failure detector.

6.7 Expert usage by reasoning phase

Experiment 5 measures each expert’s activation frequency and weight mass by phase, before/after event usage, co-activation, and top-1 transition matrices. The outputs in Table 14 and Figure 3 use current labels. Each cell reports the mean absolute activation-frequency difference over all 16×64 layer–expert cells relative to normal tokens, followed by the number of phase tokens. Token counts are not numbers of independent traces; in particular, explicit event phases remain supported by only 1–9 traces per condition.

The reliable ProcessBench first-error phase has only a 0.0104 mean absolute shift from normal expert frequencies. Comparing the five tokens after each gold first error with the five before it gives a mean absolute layer–expert change of 0.0139 over 207 events; the largest cell change is 0.0879. For the self-check conditions, five-token before/after changes at self-corrections are 0.0521 over ten GSM8K event occurrences and 0.0436 over ten MATH occurrences, but every associated trace ends incorrect and no matched-control interval is available. These are state changes around failed revision attempts, not evidence of recovery experts.

Table 15 adds the marginal-frequency baseline missing from the preliminary analysis. For each layer, the baseline is the self-transition probability under independent source and destination draws with the observed marginals, then pooled by transition count. Observed persistence is 9–10 percentage points above this baseline in

Table 15. Global top-1 expert transition persistence and a marginal-frequency independence baseline.

Condition	Transitions	Self-transition	Independence
GSM8K	5,398,816	19.79%	9.90%
GSM8K + self-check	5,456,304	19.30%	9.82%
MATH	10,396,432	29.86%	20.58%
MATH + self-check	10,390,832	29.49%	20.14%
ProcessBench	1,681,792	20.54%	10.19%

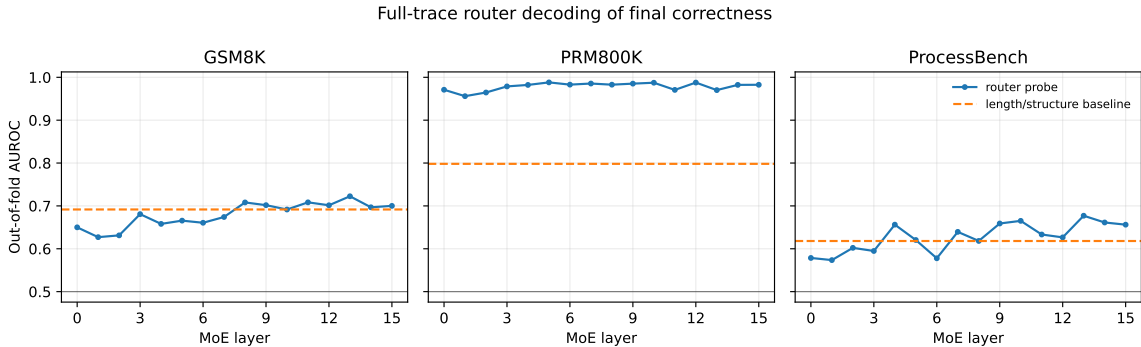


Fig. 4. Out-of-fold correctness AUROC by layer. Dashed lines use only log token count, log step count, and log character count.

every current condition. The effect is descriptive—there is no trace-level uncertainty and token identity/content are uncontrolled—but it does show that the roughly 19–30% self-transition shares are not explained solely by expert imbalance.

6.8 Legacy full-trace linear routing probes

Before the current prospective run, we trained a separate linear logistic probe at each layer using one pooled router-feature vector per trace. The 197 features comprise router-probability means and standard deviations, top-1 expert frequencies, entropy and margin statistics, and top-1 switch rate. Five-fold stratified cross-validation produces out-of-fold predictions; 95% intervals bootstrap traces. Since each trace contributes one sample, token leakage across folds is impossible. The probe pools the *complete* trace, so this is a decoding/separability test, not online prediction of a future error. These files are retained because they include the only nonempty PRM800K probe, but the GSM8K generations and labels are not identical to the current run; they are not compared numerically with Table 13.

Routing improves over the simple structure baseline by 0.031 AUROC on GSM8K, 0.190 on PRM800K correctness/first-error, 0.151 on PRM800K contradiction, and 0.059 on ProcessBench. PRM800K correctness and first-error scores are the same because, in these traces, presence of a gold first error is the complement of the final correctness label. The contradiction AUROC is high despite only 87 positives, but F1 at the AUROC-selected layer is only 0.159 under the natural class imbalance. Backtracking and self-correction probes were correctly skipped: their positive counts after relabelling are 1/1 on GSM8K, 2/1 on PRM800K, and 3/3 on ProcessBench.

Table 16. Legacy best-layer full-trace router probes. The selected-layer comparison is exploratory and is not corrected for selecting the best of 16 layers.

Dataset	Target	N	Positive	Best layer	Router AUROC [95% CI]	Balanced acc.	Structure AUROC
GSM8K	Correctness	1,319	486	13	0.7225 [0.6944, 0.7505]	0.6655	0.6917
PRM800K	Correctness	10,833	6,518	5	0.9882 [0.9858, 0.9905]	0.9760	0.7981
PRM800K	Gold first error	10,833	4,315	5	0.9882 [0.9857, 0.9904]	0.9760	0.7981
PRM800K	Contradiction	10,833	87	9	0.9082 [0.8702, 0.9385]	0.7489	0.7573
ProcessBench	Correctness	400	193	13	0.6771 [0.6244, 0.7272]	0.6356	0.6181
ProcessBench	Gold first error	400	207	13	0.6771 [0.6240, 0.7303]	0.6356	0.6181

The near-perfect PRM800K correctness AUROC (0.988) should be treated with particular caution. The probe sees the complete provided solution, the first- error target is exactly complementary to correctness in this output, and source/format artifacts can be encoded in pooled routing statistics. The current zero-input PRM800K failure prevents a prefix-only hidden/router check, so this result establishes legacy within-file decodability only.

7 Analysis

7.1 Conclusions supported so far

- (1) Hidden-state geometry strongly predicts router-distribution geometry at every layer in all five current conditions. Overall mean correlation is 0.5605–0.5663, peaks in layer 3 or 4, and is nearly invariant to self-check prompting. Correct–failed differences reverse direction across conditions.
- (2) At 207 gold ProcessBench first errors, entropy, switching, overlap, and margin do not exhibit a statistically resolved discontinuity relative to a matched position in the same trace. Simple layer-averaged routing metrics are therefore not sufficient first-error markers in the current analysis.
- (3) Self-check prompting increases explicit revision markers but does not improve final accuracy on paired GSM8K or MATH problems. Every trace labelled as a self-correction in the two self-check conditions ends incorrect.
- (4) MATH exhibits a small entropy increase at detected reasoning events, and MATH self-check backtracking spans exhibit a small margin decrease. These effects use only 9–24 traces, are uncorrected for multiple comparisons, and do not replicate as a universal multi-metric disruption.
- (5) Prospective final-incorrectness probes are usually matched or beaten by the structure baseline on GSM8K and MATH. ProcessBench combined features reach AUROC 0.702 at 75%, but its 25% future-first-error result is only 0.674 and the signal is not stable across later conditional cohorts. A robust advance detector has not been established.
- (6) Top-1 routes persist more than a marginal-frequency independence baseline in every current condition, but event-specific expert shifts remain descriptive and explicit-event trace counts remain small.
- (7) The auxiliary Qwen3.5 boundary probes show substantial sentence-split decodability of upcoming Schoenfeld episodes (selected-layer AUROC 0.835–0.934), with the strongest thresholded results for Implement, Verify, and Read. This establishes accessible prefix-state signal within familiar responses, not generalization to unseen reasoning traces.
- (8) External application of the episode probes is best interpreted as coarse phase inspection. Predictions collapse toward Implement, Read, and Monitor; Explore is never selected, 12.7% of units activate no probe, and no external gold episode labels exist. ProcessBench Verify predictions identify review-heavy passages after errors but do not identify correctness.

- (9) The current PRM800K run failed with zero input despite completion banners. Its earlier first-error stabilization and near-perfect full-trace decoding results are legacy evidence and require a clean prospective rerun.

7.2 What has not been established

The present experiments do not identify “math experts” or “failure experts”, do not show that routing changes cause reasoning errors, and do not show that backtracking forms a stable, linearly separable hidden-state class. Hidden and combined prefix probes have now been run, but their advantage over structure is inconsistent and best-layer comparisons are selection-optimistic. No t-SNE/UMAP visualization or routing intervention has been run. The geometry significance test does not supply trace-level uncertainty, event analyses are not multiplicity-corrected, and the expert transition baseline does not control token content. Correlation and linear decodability should not be read as causal evidence. The Schoenfeld boundary-probe run also does not establish response-grouped generalization, calibrated seven-way episode classification, or the cognitive states of Qwen’s own generations. It uses Qwen to encode teacher-forced DeepSeek-R1 text, shares all 38 response IDs between train and test, and selects layers on the reported test set. Its external argmax outputs have no episode ground truth and should not be called transfer accuracy.

8 Discussion

The strongest current story remains geometry-first: routing follows the evolving representation space, with a consistent layerwise signature across five dataset and prompt conditions. The anticipated universal sharp disruption at errors is not supported. ProcessBench is null at the gold first error, MATH has a selective entropy increase at heuristically detected events, and legacy PRM800K shows a small stabilization. Direction and metric therefore depend on the dataset and event definition.

The prospective runs also change the interpretation of decodability. Much of final-incorrectness prediction on generated math traces is already present in prefix length and structure; representation features add most clearly only late in ProcessBench. Future work should use paired model comparisons, prespecified layers, trace-level uncertainty, and targets defined strictly after the observed prefix. A clean PRM800K rerun is necessary before treating its unusually high legacy decoding score as a model-level phenomenon.

The auxiliary episode probes add a complementary representation-level result: the prefix before a sentence often encodes the phase likely to follow, and the signal is strongest in middle-to-late layers rather than at the final model output. The current split cannot determine whether this reflects a portable episode representation, trace-specific transition regularities, position, or formatting. A response-grouped nested evaluation should select both layer and regularization on inner folds, evaluate once on unseen responses, repeat over seeds, and report response-bootstrap intervals. Standardized or norm-matched features and simple position/previous-unit/format baselines are needed to separate hidden-state information from layer scale and trace structure.

For external use, the probe outputs should be treated as abstaining phase suggestions rather than annotations. Abbreviation- and LaTeX-aware segmentation should precede evaluation; independent one-versus-rest scores should either be calibrated on a common multiclass validation set or retain multiple labels; and a manually labelled, benchmark-stratified sample is required before reporting transfer accuracy. It would also be useful to separate cross-model recognition, as studied here, from Qwen’s own generation by running the same protocol on Qwen-generated traces.

The event-label audit also changes the experimental priority. Explicit lexical markers provide high precision but extremely low recall. Robust backtracking conclusions will require either deliberately elicited self-checking traces, manual/LLM-assisted span annotation with validation, or a benchmark with revision spans. Until then, the gold ProcessBench error analysis and outcome-based correct/failed comparisons are the safest results.

9 Conclusion

OLMoE routing is strongly and reproducibly aligned with hidden-state geometry, but aggregate metrics do not provide a dataset-invariant error signature. Self-check prompting elicits more revision language without improving paired accuracy, and prospective representation probes seldom beat simple structure baselines on generated math. ProcessBench shows a moderate late-prefix signal but no resolved discontinuity at its gold first error. Expert routes are more persistent than their marginal frequencies predict, yet event-specific expert claims remain underpowered. The auxiliary episode-judge study likewise favors conservative conclusions: context-aware light-budget GEPA with macro-F1 selection reaches 0.650 cross-validated macro-F1, but its improvement over the prior selector is small, a heavy budget is worse, and Monitor overprediction remains substantial. The separate Qwen3.5 boundary probes show strong within-split linear access to upcoming episode labels, especially for Implement, Verify, and Read, but their sentence split leaks every response across train and test and their external outputs collapse to a coarse setup/execution/review taxonomy. They therefore support prospective within-trace phase decodability, not unseen-trace generalization, automatic seven-way annotation, error detection, or claims about Qwen’s own generated cognition. The next priorities are a clean PRM800K rerun, adjudication of episode-unit artifacts, paired seed-versus-GEPA outer evaluation, response-grouped nested boundary probes with validated external labels, stronger event supervision, and only then causal routing intervention.

A Reproducibility Checklist

Table 17. Living inventory of completed and pending work.

Item	Status	Note / next action
Trace generation / ingestion	Partial	Current: GSM8K and GSM8K self-check 1,319 each; MATH and MATH self-check 1,500 each; ProcessBench 400. PRM800K is zero-byte and must be rerun.
Router/hidden extraction	Done for five conditions	All five non-PRM JSONL files validate and have complete router/hidden tensors; PRM800K current output is empty.
High-precision labels	Done/current	Current five-condition outputs use the conservative classifier; legacy PRM800K has a separate corrected-label file.
Taxonomy summary file	Done/current	results/exp1/summary.json contains the five current conditions; corrected legacy PRM800K remains in summary_relabelled.json.
ProcessBench event routing	Done/current	Gold first error has matched controls and 2,000-sample bootstrap CIs.
Generated-data event routing	Done but underpowered	Plain/self-check GSM8K and MATH are complete; event-eligible trace counts range from 2 to 24.
PRM800K event routing	Legacy only	Corrected earlier output covers 4,371 event-eligible traces; current zero-input run did not reproduce it.
Geometry-routing correlation	Done for five conditions	Overall/correct/failed and current-label backtracking outputs exist for every non-PRM condition.
Expert-event analysis	Done for five conditions	Current JSON/NPZ outputs exist; explicit event trace counts remain underpowered.
t-SNE/UMAP	Not done	Optional visualization only; stratify by layer and trace split if added.
Legacy full-trace probes	Done/legacy	Trace-grouped router and structure probes exist for GSM8K, PRM800K, and ProcessBench but are not current-run prospective results.
Hidden/combined prefix probes	Done for five conditions	Experiment 4 compares structure, router, hidden, and combined 25/50/75% prefix features.
Prospective failure prediction	Partially established	Final incorrectness is evaluated everywhere; future first error is evaluable only on ProcessBench and is not stable across conditional cohorts.
Structural/MFA analysis	Not done	Add trajectory length/curvature, reasoning structures, and optional local-geometry regions after core probes.
Routing intervention	Not done	Title-level intervention claim is not yet supported; run only after stable predictive signals are identified.
Second MoE / robustness	Not done	Run a Qwen MoE or another routing design once the corrected pipeline is frozen.
GEPA episode judge (Experiment 0a)	CV complete; validation pending	Context-aware five-fold grouped CV over 2,789 units gives 67.34% accuracy and 65.05% macro-F1 for light-budget GEPA with macro-F1 selection. Heavy is worse (66.69%, 64.37%). Review 180 flagged units, constrain decoding, evaluate seed and optimized prompts on every outer fold, and designate a genuinely unseen test set.
Schoenfeld boundary probes (Experiment 0b)	Done	Qwen3.5 GPTQ probes cover 3,087 units, seven targets, and 41 hidden-state indices; selected-layer AUROC is 0.835–0.934. External inspection covers 581 units but has no episode ground truth and never selects Explore. Grouped CV remains pending; add nested layer selection, common multiclass calibration, robust segmentation, and a manually labelled transfer set.
Pipeline status handling	Defective	The shell script prints completion after zero-input stage errors; add nonzero exits, output validation, and atomic JSONL writes.
Figures	Current after regeneration	Geometry, ProcessBench first-error, expert-phase, and legacy router-probe plots are generated by paper_figures/generate_figures.py.